ESRA-RELA++: An Explainable Saliency and Reliability-Aware Multimodal Framework for Subject-Independent Emotion Recognition

MD Nabil Thahamid Chowdhury
Computer Science and Engineering
Brac University, Dhaka, Bangladesh
thahamidnabil@gmail.com

Anisha Mahjabin Chowdhury
Computer Science and Engineering
Brac University, Dhaka, Bangladesh
anisha.mahjabin.chowdhury@g.bracu.ac.bd

***Abstract*—Automatic emotion recognition from audio-visual speech remains challenging because emotional cues are distributed across vocal prosody, facial appearance, and facial muscle dynamics. Although recent multimodal approaches have reported strong performance on the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS), performance comparisons can be affected by differences in evaluation protocol, especially whether test actors are unseen during training. RAVDESS is a validated multimodal dataset containing emotional speech and song expressions from 24 professional actors, making it suitable for subject-independent audio-visual emotion-recognition research [1].**

This paper proposes ESRA-RELA++, an explainable saliency and reliability-aware multimodal framework for subject-independent emotion recognition. The proposed system integrates four active branches: a hybrid audio branch combining frozen self-supervised speech embeddings with handcrafted acoustic and prosodic descriptors, an OpenFace-guided visual branch using emotion-salient frame selection and landmark-based face cropping, an Action Unit statistical branch, and an AU-dynamic/FACS trajectory branch. OpenFace is used to extract facial landmarks and facial Action Units, while MobileNetV3 is used as an efficient visual feature extractor [7].

The system was evaluated on the RAVDESS full audio-visual speech subset using a subject-wise 5-fold protocol across eight emotion classes. The best-performing configuration used calibrated simple-average fusion over the four active modalities, while optional LoRA audio adaptation and post-hoc specialist correction modules were evaluated as ablations. The selected model achieved 72.40% $\pm$ 3.81 accuracy, 71.94% $\pm$ 4.12 macro-F1, and 72.67% $\pm$ 3.82 unweighted average recall (UAR). Although this result does not surpass the strongest reported audio-dominant subject-wise RAVDESS result, the proposed framework provides an interpretable and modular analysis of audio, visual, AU-statistical, and AU-dynamic contributions under strict unseen-actor evaluation [2].

***Index Terms*—multimodal emotion recognition, RAVDESS, audio-visual fusion, OpenFace, Action Units, Wav2Vec2, subject-wise evaluation, reliability-aware fusion**

I. Introduction

Automatic emotion recognition aims to enable computational systems to infer human affective states from observable signals such as speech, facial expression, and body movement. In human communication, emotion is rarely expressed through a single modality only; vocal tone, facial expression, and temporal facial muscle movement often provide complementary information. For this reason, multimodal emotion recognition has become an important research direction in affective computing, human-computer interaction, social robotics, and intelligent tutoring systems. However, building a reliable multimodal system remains difficult because each modality may contain noise, missing information, speaker-specific patterns, or weak emotion cues.

The RAVDESS dataset is widely used for audio-visual emotion-recognition research because it provides a validated collection of emotional speech and song recordings from 24 professional actors. The dataset includes facial and vocal expressions in North American English and has been perceptually validated for emotion-recognition studies [1]. In this work, only the full audio-visual speech subset is used so that each sample contains both speech and facial video information. This produces 1440 full-AV speech samples across eight emotion classes: neutral, calm, happy, sad, angry, fearful, disgust, and surprised. The actor-independent subject-wise 5-fold setting is adopted so that the actors used for testing do not appear during training. This protocol is stricter than random or trial-based splitting because the model must generalize to unseen speakers and faces rather than memorizing actor-specific characteristics.

Several previous studies have investigated RAVDESS emotion recognition using audio, visual, and multimodal fusion strategies. Luna-Jiménez *et al.* proposed a multimodal approach using aural transformers and facial Action Units, reporting 86.70% accuracy on RAVDESS using subject-wise 5-fold cross-validation for eight emotion classes. Their work also emphasized the importance of high-emotional-load visual information, suggesting that visual systems could benefit from better frame selection strategies [2]. John and Kawanishi proposed an audio-video multimodal transformer architecture with audio self-attention, visual self-attention, and audio-video cross-attention branches; however, their RAVDESS experiment used first-trial sequences for training and second-trial sequences for testing, which is not equivalent to subject-wise actor-independent evaluation [3]. Therefore, reported results across papers must be interpreted carefully because differences in dataset split protocol can significantly affect the difficulty of the task.

Speech-based emotion recognition has benefited from self-supervised learning models such as wav2vec 2.0, which learn speech representations from raw audio and can be adapted to downstream speech tasks [4]. However, purely audio-dominant models may not fully explain how facial expressions and facial muscle dynamics contribute to the final decision. Similarly, visual-only approaches may suffer when facial motion is subtle, when frames are not emotionally informative, or when actor-specific facial appearance dominates the representation. To address these issues, this work combines audio, visual, and facial Action Unit information in a modular and interpretable way instead of relying on a single dominant branch.

Facial Action Units are particularly useful because they represent anatomically meaningful facial movements. The Facial Action Coding System describes visible facial movements in terms of Action Units, which can be used as interpretable indicators of facial expression behavior [6]. OpenFace provides automatic facial landmark detection, head-pose estimation, eye-gaze estimation, and facial Action Unit recognition, making it suitable for extracting structured facial behavior features from video [7]. In the proposed framework, OpenFace is used not only to extract AU statistics but also to guide emotion-salient frame selection and landmark-based face cropping. This allows the visual branch to focus on frames and facial regions that are more likely to contain expressive information.

This paper proposes ESRA-RELA++, an explainable saliency and reliability-aware multimodal framework for subject-independent RAVDESS emotion recognition. The framework contains four main active branches. First, the hybrid audio branch combines frozen self-supervised speech embeddings with handcrafted acoustic and prosodic descriptors such as MFCC, delta-MFCC, chroma, mel-spectral features, spectral descriptors, pitch-related statistics, RMS energy, and zero-crossing rate. Second, the visual branch uses OpenFace-guided emotion-salient frame selection, landmark-based face cropping, and MobileNetV3-based frame representation. MobileNetV3 is selected as an efficient convolutional visual backbone designed through hardware-aware architecture search [8]. Third, the AU-statistical branch summarizes Action Unit behavior using global statistics such as mean, standard deviation, maximum, median, interquartile range, velocity, and start-to-end change. Fourth, the AU-dynamic branch captures temporal facial behavior using segment-wise AU means, segment-wise maxima, AU velocity, peak timing, and FACS-group trajectory features.

In addition to the four main branches, optional modules were explored during experimentation. A LoRA-based audio adaptation branch was implemented as a parameter-efficient fine-tuning option. LoRA freezes the pretrained model weights and inserts trainable low-rank matrices, reducing the number of trainable parameters compared with full fine-tuning [5]. In this study, LoRA was evaluated in a fold-safe manner, meaning that LoRA models were trained only inside the training actors of each fold and never on outer-test actors. However, the LoRA configuration did not improve the final subject-wise performance and was therefore retained as an ablation rather than the selected final model. Post-hoc weak-pair specialists and a Sad-vs-rest specialist were also evaluated, but the best empirical result was obtained by calibrated simple-average fusion without LoRA and without post-hoc specialist correction.

The main contribution of this work is not to claim a new highest raw accuracy on RAVDESS. Instead, this work contributes a detailed, modular, and explainable multimodal pipeline that studies how complementary audio, visual, AU-statistical, and AU-dynamic information behave under strict subject-wise evaluation. The final selected configuration achieved 72.40% $\pm$ 3.81 accuracy, 71.94% $\pm$ 4.12 macro-F1, and 72.67% $\pm$ 3.82 UAR using four active modalities and calibrated simple-average fusion. The findings show that increasing architectural complexity does not always improve actor-independent generalization on small datasets; in this case, simpler calibrated fusion produced better performance than LoRA-enhanced or specialist-corrected variants.

The remainder of this paper is organized as follows. Section II reviews related work on RAVDESS emotion recognition, self-supervised audio representation, visual emotion analysis, and facial Action Unit modeling. Section III describes the proposed ESRA-RELA++ architecture, including preprocessing, feature extraction, and fusion. Section IV presents the experimental setup, subject-wise split protocol, and implementation details. Section V reports the results, ablation study, and class-wise error analysis. Section VI discusses the limitations and future directions, including stronger audio adaptation, improved temporal visual modeling, and more robust weak-class recognition.

II. Related Work

A. Audio-Visual Emotion Recognition and RAVDESS

Automatic emotion recognition has been widely studied using speech, facial expression, and multimodal audio-visual information. Among available datasets, the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) is one of the most commonly used benchmarks for audio-visual emotion recognition. RAVDESS contains validated emotional speech and song recordings from 24 professional actors and includes facial and vocal expressions with emotions such as calm, happy, sad, angry, fearful, disgust, and surprised, with neutral as an additional category [1]. Because RAVDESS provides both audio and visual modalities, it is suitable for evaluating whether multimodal systems can learn complementary emotional cues from voice and facial expression.

However, comparison across RAVDESS studies is not always straightforward because different works use different evaluation protocols. In random or trial-based splits, the same actor may appear in both training and testing, allowing the model to benefit from actor-specific facial or vocal characteristics. In contrast, subject-wise evaluation separates train and test actors, requiring the model to generalize to unseen identities. This makes subject-wise evaluation more difficult but more realistic for real-world emotion-recognition systems.

In this study, the subject-wise 5-fold setting is used to ensure actor-independent evaluation.

B. Prior RAVDESS Multimodal Fusion Methods

Luna-Jiménez *et al.* proposed a multimodal RAVDESS emotion-recognition system using aural transformers and facial Action Units. Their approach combined audio and visual modalities through late fusion and reported 86.70% accuracy on RAVDESS under subject-wise 5-fold cross-validation for eight emotion classes [2]. Their work is an important benchmark because it follows a subject-wise protocol and demonstrates that a strong audio branch can significantly improve final multimodal performance. They also reported that visual systems could be improved through better detection of high-emotional-load frames, which motivates the use of saliency-guided frame selection in later approaches [2].

John and Kawanishi proposed an audio-video multimodal transformer architecture for emotion recognition. Their model used three transformer branches: audio self-attention, video self-attention, and audio-video cross-attention [3]. This approach is important because it directly models cross-modal interaction between speech and facial visual information. However, reported RAVDESS results from such transformer-based systems must be interpreted carefully when the evaluation split is not subject-wise. If actor identity is shared between training and testing, the measured accuracy may not reflect generalization to unseen speakers and faces. Therefore, in this work, subject-wise evaluation is prioritized over trial-based comparison.

Compared with these prior methods, the proposed ESRA-RELA++ framework does not rely on a single audio-dominant model or a fully end-to-end transformer. Instead, it uses a modular design that separately models audio, visual appearance, AU statistics, and AU dynamics. This design allows detailed ablation analysis and makes it easier to observe which modality contributes most to the final result.

C. Speech Representation Learning for Emotion Recognition

Speech emotion recognition has benefited from self-supervised speech models such as wav2vec 2.0. Baevski *et al.* proposed wav2vec 2.0 as a framework for self-supervised speech representation learning from raw audio. The model learns latent speech representations using masked prediction and contrastive learning, and it can be fine-tuned for downstream speech tasks [4]. Because emotional speech contains important cues in pitch, rhythm, energy, and spectral variation, pretrained speech models can provide useful high-level representations for emotion recognition.

Many high-performing speech emotion systems adapt pretrained speech models through fine-tuning. However, full fine-tuning can be computationally expensive and may overfit on relatively small datasets such as RAVDESS. For this reason, the proposed framework first uses frozen self-supervised audio embeddings combined with handcrafted acoustic and prosodic descriptors. These descriptors include MFCC, delta-MFCC, chroma, mel-spectral statistics, spectral centroid, spectral bandwidth, spectral rolloff, spectral contrast, pitch-related statistics, RMS energy, and zero-crossing rate. This hybrid audio representation is designed to capture both learned speech representations and interpretable acoustic cues.

Low-Rank Adaptation (LoRA) has recently become a popular parameter-efficient fine-tuning method. LoRA freezes pretrained model weights and injects trainable low-rank decomposition matrices into transformer layers, reducing the number of trainable parameters compared with full fine-tuning [5]. In this study, LoRA was implemented as an optional audio adaptation branch. However, the LoRA branch did not improve the final subject-wise result under the current configuration, so it is reported as an ablation rather than selected as the final model. This finding supports the importance of validating parameter-efficient adaptation under strict unseen-actor evaluation rather than assuming that additional trainable parameters will always improve performance.

D. Facial Action Units and OpenFace-Based Visual Features

Facial expressions are closely related to facial muscle movement. The Facial Action Coding System (FACS) describes visible facial movements using Action Units, where each AU corresponds to individual or grouped facial muscle actions [6]. Because AUs provide interpretable facial movement information, they are useful for emotion-recognition systems that need more explainability than raw image features alone.

OpenFace is a widely used toolkit for facial behavior analysis. It supports facial landmark detection, head-pose estimation, eye-gaze estimation, and facial Action Unit recognition [7]. OpenFace 2.0 further extends this capability and provides more accurate facial landmark detection and AU recognition [7]. In this work, OpenFace is used in three ways. First, it provides AU intensity values for the AU-statistical branch. Second, it provides temporal AU information for the AU-dynamic branch. Third, it guides emotion-salient frame selection and landmark-based face cropping for the visual branch.

Previous work on RAVDESS has shown that Action Units are useful for visual emotion recognition, but static or global AU features may miss important temporal information [2]. Therefore, the proposed method uses both AU statistical features and AU-dynamic trajectory features. The AU-statistical branch summarizes global facial behavior using statistics such as mean, standard deviation, maximum, median, interquartile range, velocity, and start-to-end change. The AU-dynamic branch captures temporal facial expression patterns using segment-wise AU means, segment-wise maxima, AU velocity, peak timing, and FACS-group trajectory features.

E. Visual Feature Extraction and Saliency-Guided Frame Selection

Visual emotion recognition from video requires selecting frames that contain expressive facial information. Uniform frame sampling may include neutral or low-expression frames, especially when emotional intensity changes across time. Luna-Jiménez *et al.* noted that visual systems could benefit from detecting frames with high emotional load [2]. Motivated by this observation, the proposed visual branch uses OpenFace-derived AU intensity, AU motion, confidence, and success indicators to select emotionally salient frames.

After selecting salient frames, the proposed method crops the face using OpenFace facial landmarks. This reduces background information and focuses the visual feature extractor on the expressive facial region. MobileNetV3 is used as the visual backbone because it is an efficient convolutional architecture designed through hardware-aware neural architecture search and NetAdapt optimization [8]. The visual branch then applies temporal pooling over frame embeddings using mean, standard deviation, maximum, motion-delta, first-frame, and last-frame representations. This provides a lightweight way to include temporal information without training a heavy video transformer on a small dataset.

F. Multimodal Fusion and Reliability-Aware Modeling

Multimodal emotion recognition requires combining information from different sources. Simple late fusion combines predicted probabilities from each modality, while more complex methods use learned meta-fusion or cross-modal attention. Transformer-based methods can model cross-modal interaction, but they may require larger datasets and careful regularization to avoid overfitting [3]. In small subject-wise datasets, simpler fusion strategies may generalize better than highly complex fusion networks.

The proposed framework investigates multiple fusion strategies, including calibrated average fusion, reliability-aware out-of-fold stacking, optional LoRA audio adaptation, pair specialists, and Sad-vs-rest correction. The final selected configuration uses calibrated simple-average fusion across four active branches: hybrid audio, saliency-guided visual features, AU-statistical features, and AU-dynamic features. This configuration achieved the best observed subject-wise result in the experiments, with 72.40% accuracy, 71.94% macro-F1, and 72.67% UAR. This result suggests that, for RAVDESS under subject-wise evaluation, carefully calibrated modality averaging can outperform more complex correction modules when the dataset is limited.

G. Position of the Proposed Work

The proposed ESRA-RELA++ framework differs from previous RAVDESS approaches in several ways. Unlike audio-dominant late-fusion systems, it explicitly models four complementary sources of information: hybrid audio, saliency-guided video, AU statistics, and AU dynamics. Unlike fully end-to-end transformer-based systems, it emphasizes modularity, interpretability, subject-wise evaluation, and ablation analysis. Unlike approaches that use only static facial information, it extracts AU trajectory and FACS-group dynamic features. In addition, optional LoRA adaptation and specialist correction modules are tested experimentally, but the final configuration is selected based on subject-wise performance rather than architectural complexity.

TABLE I
CLASS DISTRIBUTION USED IN THIS STUDY

Emotion	Number of Samples
Neutral	96
Calm	192
Happy	192
Sad	192
Angry	192
Fearful	192
Disgust	192
Surprised	192
Total	1440

Thus, the main contribution of this work is not only raw accuracy improvement. Instead, it provides a structured and explainable framework for analyzing how audio, visual, AU-statistical, and AU-dynamic cues behave under strict actor-independent evaluation. This is important because emotion-recognition systems should generalize to unseen individuals, and a high score under an easier split may not reflect real-world robustness.

III. DATASET AND PREPROCESSING

A. Dataset Description

This study uses the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS), a validated multimodal emotion dataset containing facial and vocal emotional expressions from 24 professional actors. The full RAVDESS database contains acted emotional speech and song recordings across multiple modalities, including full audio-video, video-only, and audio-only files. The emotional categories include neutral, calm, happy, sad, angry, fearful, disgust, and surprised [1].

Although the complete RAVDESS database contains multiple modalities and vocal channels, this work uses only the full audio-visual speech subset. This choice was made because the proposed model is designed to combine speech audio, facial video, and facial Action Unit information from the same sample. Therefore, video-only files and song files were excluded from the main experiment. The final selected dataset contains 1440 full-AV speech samples from 24 actors and eight emotion classes. The experiment uses a strict subject-wise 5-fold split, where test actors are not used during training.

The neutral class has fewer samples because RAVDESS does not include strong-intensity neutral expressions, while the other emotions contain both normal- and strong-intensity recordings. This class imbalance was considered during evaluation by reporting not only accuracy but also macro-F1 and unweighted average recall (UAR).

B. File Selection and Label Extraction

Each RAVDESS filename follows a structured numerical format that encodes the stimulus properties, including modality, vocal channel, emotion, emotional intensity, spoken statement, repetition, and actor identity. In this study, only files satisfying modality code 01, corresponding to full audio-video, and vocal-channel code 01, corresponding to speech, were selected.

TABLE II
SUBJECT-WISE TEST ACTORS FOR EACH FOLD

Fold	Test Actors
Fold 0	2, 5, 14, 15, 16
Fold 1	3, 6, 7, 13, 18
Fold 2	10, 11, 12, 19, 20
Fold 3	8, 17, 21, 23, 24
Fold 4	1, 4, 9, 22

Thus, the preprocessing stage filtered the original local collection of approximately 4904 video files into 1440 full audio-visual speech samples. This filtering ensured that every selected sample contained both audio and video information, which is necessary for the proposed multimodal framework. The emotion identifier in the filename was converted into an eight-class label: neutral, calm, happy, sad, angry, fearful, disgust, and surprised.

Actor identity was also extracted from the filename and used only for splitting and evaluation. It was not used as a predictive feature. This is important because the objective of the study is to evaluate generalization to unseen actors rather than memorization of actor-specific facial or vocal patterns.

C. Subject-Wise 5-Fold Evaluation Protocol

A subject-wise 5-fold protocol was used to evaluate the model. In each fold, a group of actors was held out for testing, while the remaining actors were used for training. This setup is more difficult than random splitting because the model must classify emotions from actors it has never seen during training.

This protocol follows an actor-independent evaluation strategy. Four folds contain five test actors, while one fold contains four test actors. The final results are reported as mean $\pm$ standard deviation across the five folds.

D. OpenFace Preprocessing

For the visual and facial Action Unit branches, each selected video was processed using OpenFace. OpenFace is an open-source facial behavior analysis toolkit capable of facial landmark detection, head-pose estimation, eye-gaze estimation, and facial Action Unit recognition [7].

OpenFace produced one CSV file for each video. These CSV files were used to extract facial landmark coordinates, frame-level OpenFace confidence, frame-level tracking success, facial Action Unit intensity values, and temporal Action Unit movement information. In the final experiment, OpenFace coverage was complete: 1440 of 1440 selected videos had corresponding OpenFace CSV files, giving 100% coverage.

E. Audio Preprocessing

The audio stream was extracted from each full-AV speech video and resampled to 16,000 Hz. Each signal was represented using a fixed duration of 5.5 seconds. If an audio signal was shorter than this duration, zero-padding was applied. If it was longer, it was truncated to the fixed length. This fixed-duration representation ensured that all samples had a consistent input format.

Two types of audio features were extracted. First, a frozen self-supervised speech model was used to obtain high-level speech embeddings. In the final selected configuration, the audio branch used `facebook/wav2vec2-base-960h`. Second, handcrafted acoustic and prosodic descriptors were extracted to capture interpretable speech-emotion cues. These included MFCC, delta-MFCC, chroma features, mel-spectrogram statistics, spectral centroid, spectral bandwidth, spectral rolloff, spectral contrast, pitch/F0 statistics, RMS energy, and zero-crossing rate. The final audio feature vector was formed by concatenating the frozen SSL speech embedding with the handcrafted acoustic-prosodic feature vector.

F. Video Preprocessing

The visual branch used the original video frames together with OpenFace information. Instead of uniformly selecting frames only, OpenFace-derived facial behavior scores were used to identify emotionally informative frames. The saliency score was based on Action Unit intensity, Action Unit movement or change, OpenFace confidence, and OpenFace tracking success.

For each video, the model selected 10 representative frames from a saliency candidate pool of 24 frames. After frame selection, the face region was cropped using the 68 OpenFace facial landmarks. This step reduced background information and focused the visual representation on the facial expression region. The cropped frames were then passed through a MobileNetV3-based feature extractor. MobileNetV3 is an efficient convolutional architecture designed for lightweight visual representation learning [8].

The final video representation used temporal pooling over frame embeddings, including mean embedding, standard deviation embedding, maximum embedding, frame-difference or motion embedding, first-frame embedding, and last-frame embedding. This allowed the video branch to preserve both appearance and lightweight temporal information.

G. Action Unit Statistical Preprocessing

Facial Action Units provide interpretable information about facial muscle movement. The Facial Action Coding System represents facial behavior through Action Units, which are useful for explaining emotional expression patterns [6].

The AU statistical branch used the OpenFace AU intensity columns. For each AU, mean, standard deviation, maximum, minimum, median, interquartile range, average velocity, and start-to-end change were extracted. These statistics summarize the overall facial muscle behavior across the full video clip.

H. AU-Dynamic Preprocessing

In addition to the AU-statistical branch, this work introduces an AU-dynamic branch to capture temporal facial muscle behavior from the OpenFace Action Unit sequence. The motivation for this branch is that emotional facial expression is not only defined by the overall intensity of Action Units, but also by how those Action Units change over time. Previous AU-based studies have shown that collapsing frame-level AU

TABLE III
FACS-INSPIRED AU GROUPS USED FOR AU-DYNAMIC FEATURES

Group	Action Units
Brow	AU01, AU02, AU04
Eye	AU05, AU06, AU07, AU45
Nose	AU09, AU10
Mouth	AU12, AU14, AU15, AU17, AU20, AU23, AU25, AU26

sequences into a single average vector is simple and robust, but it may ignore temporal ordering and reduce discriminative information in sequential facial behavior.

Let the OpenFace AU sequence for a video be represented as

$$\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_T], \tag{1}$$

where T is the number of OpenFace frames and each $\mathbf{u}_t \in \mathbb{R}^{17}$ contains the 17 AU intensity values used in this study: AU01, AU02, AU04, AU05, AU06, AU07, AU09, AU10, AU12, AU14, AU15, AU17, AU20, AU23, AU25, AU26, and AU45.

The AU sequence is divided into five equal temporal segments, S_1, S_2, S_3, S_4, and S_5, where each segment contains a consecutive portion of the full AU sequence. If the total number of frames is not exactly divisible by five, the boundary indices are generated using linear spacing across the full sequence. For each segment S_k, the segment-wise AU mean and maximum are calculated:

$$\boldsymbol{\mu}_k = \frac{1}{|S_k|} \sum_{t \in S_k} \mathbf{u}_t, \tag{2}$$

$$\mathbf{m}_k = \max_{t \in S_k} (\mathbf{u}_t), \tag{3}$$

where $\boldsymbol{\mu}_k$ and $\mathbf{m}_k$ are 17-dimensional vectors. The segment-wise means and maxima preserve coarse temporal structure by representing how facial muscle activity changes from the beginning to the end of the clip.

The AU velocity is computed from the first-order temporal difference:

$$\Delta \mathbf{u}_t = \mathbf{u}_t - \mathbf{u}_{t-1}. \tag{4}$$

For each AU, the mean absolute velocity and maximum absolute velocity are calculated as

$$\mathbf{v}_{mean} = \frac{1}{T-1} \sum_{t=2}^{T} |\Delta \mathbf{u}_t|, \tag{5}$$

$$\mathbf{v}_{max} = \max_{t=2}^{T} |\Delta \mathbf{u}_t|. \tag{6}$$

The normalized peak timing is also extracted for each AU. For the j-th AU, the peak timing is defined as

$$p_j = \frac{\arg\max_t u_{t,j}}{T-1}, \tag{7}$$

where $p_j \in [0, 1]$. This feature indicates whether the strongest activation of a particular AU occurs near the beginning, middle, or end of the expression.

To add interpretable facial-region dynamics, four FACS-inspired AU groups are defined in Table III.

For each group, a group-level temporal trajectory is computed by averaging the AU intensities belonging to that group:

$$g_q(t) = \frac{1}{|G_q|} \sum_{j \in G_q} u_{t,j}, \tag{8}$$

where G_q is one of the four AU groups. From each group trajectory, five descriptors are computed:

$$[\mu(g_q), \sigma(g_q), \max(g_q), g_q(T) - g_q(1), \mu(|\Delta g_q|)]. \tag{9}$$

These values describe the average activation, variation, maximum activation, start-to-end change, and average motion of each facial region. In the implemented system, the FACS-group trajectory vector keeps 18 group-level values from these group descriptors to maintain a fixed feature size consistent with the experimental pipeline. The V15.1 implementation defines the AU-dynamic branch as part of the OpenFace AU feature pipeline and saves it as a separate modality for fusion and ablation analysis.

The final AU-dynamic feature vector is therefore

$$\begin{aligned} \mathbf{x}_{AU\text{-}dyn} = [&\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_5, \mathbf{m}_1, \ldots, \mathbf{m}_5, \mathbf{v}_{mean}, \\ &\mathbf{v}_{max}, \mathbf{p}, \mathbf{g}_{FACS}, \mathbf{r}_{AU\text{-}dyn}], \end{aligned} \tag{10}$$

where $\mathbf{g}_{FACS}$ represents the retained FACS-group trajectory descriptors and $\mathbf{r}_{AU\text{-}dyn}$ contains AU-dynamic reliability information, including AU-dynamic availability, OpenFace confidence, OpenFace success rate, and AU-dynamic feature norm. This branch allows the model to capture temporal facial expression behavior without training a heavy sequence model, which is useful for the relatively small subject-wise RAVDESS setting.

I. Reliability Feature Construction

A reliability vector was also constructed for each sample. This vector did not directly represent emotion, but described the quality or reliability of each modality. It included audio RMS energy, audio zero-crossing rate, audio feature norm, number of video frames used, whether saliency information was available, video feature norm, AU availability, AU confidence, AU success rate, AU-dynamic availability, AU-dynamic confidence, and AU-dynamic success rate.

The purpose of these reliability features was to support modality analysis and fusion decisions. For example, if OpenFace confidence was low for a sample, AU-based features might be less reliable.

J. Metadata Leakage Prevention

RAVDESS filenames contain metadata such as emotional intensity, spoken statement, and repetition number. These values can sometimes act as shortcuts if used incorrectly. Therefore, this study excluded emotional intensity, statement ID, and repetition ID from the default feature representation.

Only the emotion label was used as the target class, and actor identity was used only for subject-wise splitting. This helped keep the evaluation fair and reduced the risk of metadata leakage. The final configuration confirms that metadata covariates were excluded by default.

K. Final Preprocessed Branches

After preprocessing, each sample had four active feature branches in the final selected model: the hybrid audio branch, saliency-guided video branch, AU statistical branch, and AU-dynamic branch.

The optional LoRA audio branch, pair-specialist correction, gender-specialist correction, and Sad-vs-rest correction were tested during experimentation but were not used in the final selected configuration because the best performance was obtained without them. The final selected setup used calibrated simple-average fusion across the four active branches and achieved 72.40% ± 3.81 accuracy, 71.94% ± 4.12 macro-F1, and 72.67% ± 3.82 UAR under subject-wise 5-fold evaluation.

IV. Methodology

A. Overview of the Proposed Framework

This study proposes ESRA-RELA++, an explainable saliency and reliability-aware multimodal framework for subject-independent emotion recognition using the RAVDESS full audio-visual speech subset. The proposed system is designed to combine complementary emotional cues from four active branches: hybrid audio, saliency-guided facial video, Action Unit statistical features, and Action Unit dynamic trajectory features. The final selected configuration uses calibrated simple-average fusion over these four active modalities, while optional modules such as LoRA audio adaptation and post-hoc specialist correction are evaluated separately as ablation experiments.

The complete framework begins with a RAVDESS full-AV speech sample and processes it through four parallel branches. The audio branch combines a frozen wav2vec2 SSL embedding with handcrafted acoustic and prosodic features. The video branch applies OpenFace-guided saliency frame selection, landmark-based face cropping, and MobileNetV3 temporal visual embedding. The AU statistical branch extracts global Action Unit statistics, while the AU-dynamic branch extracts AU trajectory, velocity, peak timing, and FACS-group dynamics. Modality-specific classifiers then produce posterior probabilities, posterior calibration is applied, and calibrated average fusion generates the final emotion prediction. The final output is one of eight emotion classes: neutral, calm, happy, sad, angry, fearful, disgust, and surprised.

B. Hybrid Audio Branch

The audio branch is designed to capture both high-level learned speech representations and interpretable acoustic-emotional cues. First, the audio signal is resampled to 16 kHz and normalized to a fixed duration of 5.5 seconds. Shorter clips are zero-padded, while longer clips are truncated. This ensures that each sample has a consistent input duration.

The first part of the audio branch uses a frozen self-supervised speech representation model. In the final selected configuration, the model uses `facebook/wav2vec2-base-960h` to extract high-level speech embeddings. wav2vec 2.0 is a self-supervised speech representation model that learns useful speech features from raw audio and has been widely used for downstream speech tasks [4]. Since the model is used as a frozen feature extractor in the final selected configuration, its parameters are not updated during training.

For each audio clip, the hidden-state sequence from wav2vec2 is pooled using

$$\mathbf{a}_{ssl} = [\mathrm{mean}(\mathbf{H}), \mathrm{std}(\mathbf{H}), \max(\mathbf{H})], \tag{11}$$

where $\mathbf{H}$ represents the frame-level hidden-state sequence from the pretrained speech model.

In addition to SSL embeddings, handcrafted acoustic and prosodic descriptors are extracted. These include MFCC, delta-MFCC, chroma, mel-spectrogram statistics, spectral centroid, spectral bandwidth, spectral rolloff, spectral contrast, pitch/F0 statistics, RMS energy, and zero-crossing rate. The final audio representation is obtained by concatenating the SSL embedding and handcrafted feature vector:

$$\mathbf{x}_{audio} = [\mathbf{a}_{ssl}; \mathbf{a}_{hand}]. \tag{12}$$

This hybrid design allows the audio branch to preserve both learned speech representations and interpretable prosodic cues.

C. Optional LoRA Audio Adaptation Branch

An optional LoRA audio branch was also implemented. LoRA, or Low-Rank Adaptation, is a parameter-efficient fine-tuning method that freezes pretrained model weights and injects trainable low-rank matrices into selected layers [5]. In this work, LoRA was added as a separate optional audio modality rather than replacing the frozen audio branch.

When enabled, the LoRA branch trains a wav2vec2-based sequence-classification model inside each training fold. The LoRA branch is designed to be fold-safe: the outer test actors are never used during LoRA training. For each outer fold, LoRA models are trained only on the outer-train actors, and inner out-of-fold predictions are produced for meta-level training. The V15.1 implementation explicitly treats LoRA as an optional fold-safe modality and excludes outer-test actors during adaptation.

However, the experimental results showed that the LoRA branch did not improve the final subject-wise performance under the current configuration. Therefore, LoRA was not used in the final selected model and was reported as an ablation study instead.

D. Saliency-Guided Facial Video Branch

The video branch extracts facial appearance and lightweight temporal visual information from the video stream. Instead of selecting frames uniformly only, OpenFace-derived facial behavior information is used to identify more emotionally informative frames. OpenFace provides facial landmarks, tracking confidence, success indicators, and facial Action Unit information [7].

For each video, the frame saliency score is computed from AU intensity, AU temporal change, OpenFace confidence, and OpenFace tracking success. Frames with stronger AU activation and reliable OpenFace tracking are considered more emotionally salient. From a candidate pool of frames, the model selects a fixed number of representative frames. In the final setup, 10 frames are selected from a saliency candidate pool of 24 frames. After frame selection, the face region is cropped using the 68 OpenFace facial landmarks. This reduces background noise and focuses the visual representation on the expressive facial region.

Each selected face crop is then passed through MobileNetV3, an efficient convolutional neural network designed for lightweight visual representation learning [8]. The frame embeddings are pooled using temporal statistics:

$$\mathbf{x}_{video} = [\mu(\mathbf{V}), \sigma(\mathbf{V}), \max(\mathbf{V}), \Delta(\mathbf{V}), \mathbf{V}_1, \mathbf{V}_T], \tag{13}$$

where $\mathbf{V}$ is the sequence of frame embeddings, μ is the mean, σ is the standard deviation, Δ represents average frame-to-frame change, $\mathbf{V}_1$ is the first selected frame embedding, and $\mathbf{V}_T$ is the last selected frame embedding. This branch captures both static facial appearance and lightweight temporal variation.

E. Action Unit Statistical Branch

Facial Action Units provide interpretable information about facial muscle movement. The Facial Action Coding System represents facial behavior through Action Units, which are anatomically meaningful facial movement descriptors [6]. Since emotions are often expressed through facial muscle activation, AU-based features provide an interpretable visual-emotion representation.

The AU statistical branch uses OpenFace AU intensity columns. For each Action Unit, mean, standard deviation, maximum, minimum, median, interquartile range, average velocity, and start-to-end change are computed across the full video clip. For an Action Unit sequence $\mathbf{u}$, the AU statistical representation can be written as

$$\begin{aligned}\mathbf{s}_{AU} = [&\mu(\mathbf{u}), \sigma(\mathbf{u}), \max(\mathbf{u}), \min(\mathbf{u}), \mathrm{median}(\mathbf{u}), \\ &\mathrm{IQR}(\mathbf{u}), \mu(|\Delta\mathbf{u}|), u_T - u_1].\end{aligned} \tag{14}$$

The statistics from all selected AUs are concatenated to form the AU statistical feature vector:

$$\mathbf{x}_{AU\text{-}stat} = [\mathbf{s}_{AU_1}; \mathbf{s}_{AU_2}; \ldots; \mathbf{s}_{AU_n}]. \tag{15}$$

F. AU-Dynamic Trajectory Branch

The AU statistical branch summarizes global facial activity, but it may not fully capture how facial expressions evolve over time. Therefore, an additional AU-dynamic branch is introduced to model temporal facial muscle behavior. The AU-dynamic branch extracts segment-wise AU means, segment-wise AU maxima, mean AU velocity, maximum AU velocity, normalized AU peak timing, and FACS-group trajectory features. Each AU sequence is divided into temporal segments. For each segment, the mean and maximum activation are computed. This allows the model to observe how AU activity changes during the emotional expression.

For AU sequence $\mathbf{u}$, the dynamic representation includes

$$\begin{aligned}\mathbf{d}_{AU} = [&\mu(\mathbf{u}_{seg1}), \ldots, \mu(\mathbf{u}_{segk}), \max(\mathbf{u}_{seg1}), \ldots, \\ &\max(\mathbf{u}_{segk}), \mu(|\Delta\mathbf{u}|), \max(|\Delta\mathbf{u}|), t_{peak}],\end{aligned} \tag{16}$$

where t_{peak} is the normalized time position of the maximum AU activation. The branch also computes group-level FACS features from interpretable facial regions such as the brow, eye, nose, and mouth. The final AU-dynamic vector is

$$\mathbf{x}_{AU\text{-}dyn} = [\mathbf{d}_{AU_1}; \mathbf{d}_{AU_2}; \ldots; \mathbf{g}_{FACS}], \tag{17}$$

where $\mathbf{g}_{FACS}$ represents the group-level facial trajectory features.

G. Reliability Feature Construction

A reliability vector is created for each sample to describe the quality and availability of each modality. This vector is not an emotion descriptor by itself; instead, it provides information about how reliable each extracted branch may be. The reliability vector includes audio RMS energy, audio zero-crossing rate, audio feature norm, number of video frames used, video saliency availability, video feature norm, AU availability, AU confidence, AU success rate, AU-dynamic availability, AU-dynamic confidence, and AU-dynamic success rate.

The reliability vector can be represented as

$$\mathbf{r} = [r_{audio}, r_{video}, r_{AU}, r_{AU\text{-}dyn}], \tag{18}$$

where each component contains modality-specific quality indicators. Although the code supports reliability-aware meta-fusion, the final selected model uses calibrated simple-average fusion because it gave the best observed subject-wise performance.

H. Modality-Specific Classifiers

Each active branch is used to train a separate modality-specific classifier. The active branches in the final selected model are the hybrid audio branch, saliency-guided video branch, AU statistical branch, and AU-dynamic branch. For each branch, the feature vector is standardized using `StandardScaler`. Principal Component Analysis (PCA) is applied when the feature dimensionality is high. A logistic regression classifier with balanced class weighting is then trained for each modality.

For modality m, the classifier outputs an eight-class posterior probability vector:

$$\mathbf{p}^{(m)} = [p_1^{(m)}, p_2^{(m)}, \ldots, p_8^{(m)}], \tag{19}$$

where each value corresponds to one emotion class.

I. Subject-Wise Out-of-Fold Training

To avoid leakage during model selection and probability generation, the training process uses actor-grouped out-of-fold prediction. Within each outer training fold, inner GroupKFold splitting is applied using actor IDs as groups. This means that predictions used for fusion are generated from models that did not train on the same actor group.

For each modality, the model produces out-of-fold training probabilities and outer-test probabilities. This is important because the fusion stage should not learn from overconfident predictions produced on samples already seen during training.

J. Posterior Temperature Calibration

The modality classifiers can produce overconfident probability distributions. Therefore, posterior temperature calibration is applied to smooth or sharpen modality probabilities before fusion. Since the branch classifiers output probabilities rather than neural logits, calibration is applied by raising probabilities to the power of $1/T$:

$$\tilde{p}_i = \frac{p_i^{1/T}}{\sum_{j=1}^{C} p_j^{1/T}}, \tag{20}$$

where T is the temperature value and $C = 8$ is the number of emotion classes. The temperature is selected using out-of-fold probabilities from the training fold. This produces calibrated posterior probabilities for each modality.

K. Fusion Strategy

Several fusion strategies were tested during experimentation, including reliability-aware meta-fusion, weighted dual fusion, LoRA-enhanced fusion, post-hoc specialists, and simple average fusion. The final selected configuration used calibrated simple-average fusion because it achieved the best subject-wise performance.

For the final model, the calibrated probability vectors from the four active modalities are averaged:

$$\mathbf{p}_{final} = \frac{1}{4}(\tilde{\mathbf{p}}_{audio} + \tilde{\mathbf{p}}_{video} + \tilde{\mathbf{p}}_{AU\text{-}stat} + \tilde{\mathbf{p}}_{AU\text{-}dyn}). \tag{21}$$

The predicted class is

$$\hat{y} = \arg\max_{c} \mathbf{p}_{final,c}. \tag{22}$$

This simple fusion strategy outperformed the more complex LoRA-enhanced and specialist-corrected variants in the final experiments. The final selected simple-average no-LoRA configuration achieved 72.40% $\pm$ 3.81 accuracy, 71.94% $\pm$ 4.12 macro-F1, and 72.67% $\pm$ 3.82 UAR.

L. Optional Specialist Modules

The framework also includes optional post-hoc specialist modules. These were designed to target common emotion-confusion pairs, such as Fearful vs. Surprised, Sad vs. Disgust, Sad vs. Fearful, Angry vs. Disgust, and Neutral vs. Calm. A separate Sad-vs-rest specialist was also implemented because Sad was one of the weakest classes in earlier experiments.

However, these specialist modules were disabled in the final selected configuration because they did not improve the best subject-wise result. The final summary confirms that `enable_specialists=false` and `enable_sad_specialist=false` in the best-performing run.

M. Final Selected Configuration

The final selected configuration uses the audio branch, video branch, AU statistical branch, and AU-dynamic branch. The LoRA branch, pair specialists, and Sad specialist are disabled. Fusion is performed using calibrated simple-average fusion. The final model is therefore a four-branch multimodal system using calibrated average probability fusion. This configuration was selected because it provided the highest observed subject-wise performance among the tested variants.

N. Evaluation Metrics

The model is evaluated using three metrics: accuracy, macro-F1, and unweighted average recall (UAR). Accuracy measures the overall proportion of correct predictions:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}. \tag{23}$$

Macro-F1 computes the average F1-score across all classes:

$$MacroF1 = \frac{1}{C}\sum_{i=1}^{C} F1_i. \tag{24}$$

UAR computes the average recall across all emotion classes:

$$UAR = \frac{1}{C}\sum_{i=1}^{C} Recall_i. \tag{25}$$

UAR is especially important for RAVDESS because the neutral class has fewer samples than the other emotion classes.

O. Methodological Summary

The proposed methodology is designed to balance performance, interpretability, and subject-independent evaluation. Instead of relying on only one dominant branch, ESRA-RELA++ extracts four complementary representations: audio, facial video, AU statistics, and AU dynamics. The final experiments show that calibrated average fusion of these four branches provides better generalization than more complex LoRA or specialist-enhanced variants under subject-wise 5-fold evaluation.

V. Evaluation

A. Experimental Setup

The proposed ESRA-RELA++ framework was evaluated on the RAVDESS full audio-visual speech subset using a strict subject-wise 5-fold protocol. The local dataset contained 4904 video files before filtering; however, only full audio-video speech samples were selected for the main experiment. After filtering, the final dataset contained 1440 samples, 24 actors, and eight emotion classes. The final selected configuration used four active branches: hybrid audio, saliency-guided video, AU statistical features, and AU-dynamic trajectory features. The LoRA audio branch, pair specialists, and Sad-vs-rest specialist were evaluated separately but disabled in the best-performing configuration.

The final selected configuration enabled the audio branch, video branch, AU statistical branch, and AU-dynamic branch. The LoRA audio branch, pair specialists, and Sad-vs-rest specialist were disabled. The fusion strategy was calibrated simple-average fusion. The system used `facebook/wav2vec2-base-960h` as the frozen self-supervised audio feature extractor, OpenFace CSV files for facial landmarks and Action Unit information, MobileNetV3

TABLE IV
SUBJECT-WISE EVALUATION SPLIT

Fold	Test Actors
Fold 0	2, 5, 14, 15, 16
Fold 1	3, 6, 7, 13, 18
Fold 2	10, 11, 12, 19, 20
Fold 3	8, 17, 21, 23, 24
Fold 4	1, 4, 9, 22

for visual feature extraction, and logistic regression classifiers with balanced class weighting for the modality-specific classifiers. The final fusion was performed by averaging the calibrated posterior probabilities from the four active branches.

B. Evaluation Protocol

A subject-wise 5-fold evaluation protocol was used. In this setting, actors in the test fold were completely excluded from training. This means the model was evaluated on unseen actors rather than on samples from actors already observed during training. The actor split used in the experiment is shown in Table IV.

Four folds used five test actors, while one fold used four test actors. Therefore, final results were reported as mean $\pm$ standard deviation across all five folds. This protocol was selected because it provides a stricter evaluation of generalization to unseen speakers and faces.

C. Evaluation Metrics

Three metrics were used to evaluate performance: accuracy, macro-F1, and unweighted average recall (UAR). Accuracy measures the overall proportion of correctly classified samples:

$$Accuracy = \frac{N_{correct}}{N_{total}}. \tag{26}$$

Macro-F1 computes the average F1-score across all emotion classes:

$$MacroF1 = \frac{1}{C}\sum_{i=1}^{C} F1_i, \tag{27}$$

where C is the number of emotion classes. UAR measures the average recall across all emotion classes:

$$UAR = \frac{1}{C}\sum_{i=1}^{C} Recall_i. \tag{28}$$

UAR is particularly important in this experiment because the neutral class has fewer samples than the other emotion classes. Therefore, reporting UAR and macro-F1 gives a more balanced view of class-wise performance than accuracy alone.

D. Final Fold-Wise Results

The best-performing configuration was the calibrated simple-average no-LoRA model. The fold-wise results are shown in Table V.

These results show that the proposed four-branch calibrated fusion model provides stable subject-wise performance across different actor groups. Fold 1 achieved the highest performance, while Fold 3 was the most difficult fold. The performance difference across folds suggests that actor-specific variation remains an important challenge in subject-independent emotion recognition.

E. Class-Wise Performance

The class-wise classification report for the final selected configuration is shown in Table VII.

The strongest classes were Happy, Disgust, Calm, and Angry. These emotions generally contain more distinct facial and vocal expression patterns, which helped the multimodal fusion model classify them more accurately. The weakest classes were Sad, Surprised, and Fearful. Sad was especially difficult because it was often confused with other low-arousal or ambiguous emotions. Surprised and Fearful also remained challenging because both classes can share similar high-arousal facial cues, such as widened eyes and strong mouth movement.

F. Ablation Study

Ablation experiments were conducted to evaluate the contribution of each modality and to determine whether additional modules such as LoRA adaptation, meta-fusion, and specialist correction improved performance. The final selected model used four active branches: hybrid audio, saliency-guided video, AU-statistical features, and AU-dynamic trajectory features. LoRA, pair specialists, gender specialists, and Sad-vs-rest specialist correction were tested separately but were not included in the final selected model because they did not improve the best subject-wise result. The final selected configuration uses `use_audio=true`, `use_video=true`, `use_au=true`, `use_au_dynamic=true`, `use_lora_audio=false`, `enable_specialists=false`, `enable_sad_specialist=false`, and calibrated simple-average fusion.

1) Individual Modality Test Performance: Table VIII reports the individual modality performance under the same subject-wise 5-fold setting. These results show how each branch performs before fusion.

The individual modality results show that the AU-based branches were stronger than the standalone audio and video branches. The AU-statistical branch achieved 58.02% UAR, while the AU-dynamic branch achieved 58.20% UAR. This indicates that structured facial muscle information from OpenFace contributed strongly to the model. The audio-only branch achieved 48.30% UAR, and the video-only branch achieved 43.33% UAR, showing that neither audio nor facial appearance alone was sufficient for strong subject-wise performance in the current configuration.

The calibrated simple-average fusion achieved 72.40% accuracy, 71.94% macro-F1, and 72.67% UAR, outperforming every individual branch by a large margin. This confirms that the four branches provide complementary information. The fusion gain is especially important because it shows that even

TABLE V
FOLD-WISE PERFORMANCE OF THE FINAL SELECTED CONFIGURATION

Fold	Test Actors	Accuracy	Macro-F1	UAR	Uncertain Samples
Fold 0	2, 5, 14, 15, 16	73.00%	72.58%	74.06%	58
Fold 1	3, 6, 7, 13, 18	77.33%	76.14%	77.81%	60
Fold 2	10, 11, 12, 19, 20	70.33%	69.43%	70.00%	64
Fold 3	8, 17, 21, 23, 24	66.33%	65.40%	66.88%	66
Fold 4	1, 4, 9, 22	75.00%	76.13%	74.61%	42

TABLE VI
AVERAGE PERFORMANCE OF THE FINAL SELECTED CONFIGURATION

Metric	Result
Accuracy	72.40% ± 3.81
Macro-F1	71.94% ± 4.12
UAR	72.67% ± 3.82

TABLE VII
CLASS-WISE PERFORMANCE OF THE FINAL SELECTED CONFIGURATION

Emotion	Precision	Recall	F1-score	Support
Neutral	71.15%	77.08%	74.00%	96
Calm	75.24%	80.73%	77.89%	192
Happy	83.50%	86.98%	85.20%	192
Sad	61.49%	51.56%	56.09%	192
Angry	77.04%	78.65%	77.84%	192
Fearful	62.24%	63.54%	62.89%	192
Disgust	80.19%	86.46%	83.21%	192
Surprised	62.94%	55.73%	59.12%	192

when individual modalities are moderate, their calibrated combination can produce substantially stronger actor-independent performance.

2) Out-of-Fold Modality Reliability: In addition to test-fold modality performance, the framework records out-of-fold UAR values for each modality. These values are computed only from training-fold out-of-fold predictions and are used to analyze how reliable each modality appears before final test prediction. The saved output file `r08_modality_oof_uar.csv` reports the average OOF-UAR values shown in Table IX.

The OOF-UAR values support the same conclusion as the test-fold ablation: the AU-statistical and AU-dynamic branches are the most reliable individual modalities in this experimental setup. The audio branch provides useful complementary information, but it is weaker than highly optimized audio-dominant systems reported in prior work. The video branch is the weakest standalone branch, but it still contributes additional visual information when fused with the other modalities.

3) Fusion-Level Comparison: Several fusion-level configurations were also evaluated. The main comparison is shown in Table X.

The fusion-level ablation shows that the final selected calibrated simple-average configuration performed better than the more complex meta-fusion and specialist-corrected versions. The LoRA-enhanced version also performed worse than the selected no-LoRA model. This does not imply that LoRA is ineffective in general; rather, under the current subject-wise RAVDESS setting and hyperparameter configuration, the LoRA audio branch introduced weaker or noisier probability estimates and did not improve final generalization.

These results show that the best-performing configuration was not the most complex architecture. The most effective setting was a stable calibrated fusion of complementary modalities. This is an important finding because it demonstrates that, in small actor-independent datasets, additional adaptation or post-hoc correction modules must be validated carefully rather than assumed to improve performance.

The ablation results provide an important methodological insight. Although the framework supports reliability-aware meta-fusion, LoRA audio adaptation, weak-pair specialists, and Sad-vs-rest correction, the best observed subject-wise performance was obtained by calibrated simple-average fusion. This indicates that the four active branches provide complementary but individually moderate signals, and that stable probability averaging generalizes better than additional correction layers in the current dataset setting. Therefore, the final model selection was based on empirical subject-wise performance rather than architectural complexity.

G. Uncertainty Analysis

The model also recorded low-confidence predictions using an uncertainty threshold. In the final selected configuration, uncertainty logging did not change the predicted class; it only marked samples where the maximum predicted probability was below the threshold. The fold-wise number of uncertain samples ranged from 42 to 66. Fold 3 had the highest number of uncertain samples and also the lowest fold accuracy, suggesting that uncertainty logging is useful for identifying difficult actor groups or ambiguous emotion samples.

The uncertainty module is therefore useful for post-hoc analysis, even though it was not used as an abstention mechanism. In future work, this uncertainty information could be used to design rejection-based classification or human-in-the-loop review for low-confidence predictions.

H. Discussion of Main Findings

The evaluation provides several important findings. First, the best result was obtained using four active modalities: audio, video, AU statistics, and AU dynamics. This confirms that emotional information is distributed across vocal, visual, and facial muscle movement cues. Second, AU-based features were highly useful. The AU statistical and AU-dynamic branches helped capture interpretable facial muscle behavior and temporal facial expression changes. This supports the importance of OpenFace-based structured facial features in emotion recognition.

TABLE VIII
INDIVIDUAL MODALITY AND FUSION ABLATION RESULTS

Model / Branch	Accuracy	Macro-F1	UAR
Audio only	48.70% ± 3.98	48.29% ± 3.42	48.30% ± 4.13
Video only	43.87% ± 6.05	41.99% ± 7.19	43.33% ± 6.25
AU-statistical only	57.88% ± 3.48	57.29% ± 2.64	58.02% ± 2.69
AU-dynamic only	57.92% ± 3.25	56.91% ± 2.43	58.20% ± 3.12
Calibrated average fusion	72.40% ± 4.26	71.94% ± 4.60	72.67% ± 4.27

TABLE IX
MEAN OUT-OF-FOLD UAR BY MODALITY

Modality	Mean OOF-UAR	Standard Deviation
Audio	48.76%	1.39
Video	42.25%	1.11
AU-statistical	56.29%	2.51
AU-dynamic	55.67%	1.10

Third, the LoRA audio branch did not improve final performance. Although LoRA was trained in a fold-safe manner, it introduced additional noisy probabilities under the current setup. This suggests that parameter-efficient fine-tuning does not automatically improve subject-wise emotion recognition when the dataset is small. Fourth, post-hoc specialist correction did not improve the best result. The pair specialists and Sad-vs-rest specialist were designed to correct difficult emotion pairs, but the final selected configuration performed better when these modules were disabled. This indicates that specialist correction may over-correct predictions when the base modality fusion is already stable.

Finally, the best-performing result came from a simpler calibrated fusion strategy rather than the most complex architecture. This is an important methodological finding: in subject-independent emotion recognition, robust generalization may benefit more from stable complementary modalities than from excessive post-processing complexity.

I. Comparison with Prior Work

The proposed method does not exceed the strongest reported subject-wise RAVDESS accuracy from prior audio-dominant work. However, the contribution of this study is different. Instead of only optimizing a single strong audio model, ESRA-RELA++ provides a modular and explainable analysis of multiple modalities: hybrid audio, saliency-guided video, AU statistics, and AU dynamics.

Some prior works report higher RAVDESS accuracy under different evaluation protocols, such as trial-based or non-subject-independent splits. These results are not directly comparable to the subject-wise setting because actor identity may be shared between training and testing. In contrast, this study uses actor-independent evaluation, where test actors are unseen during training. Therefore, the reported result reflects a stricter generalization setting.

The final selected configuration achieved 72.40% ± 3.81 accuracy, 71.94% ± 4.12 macro-F1, and 72.67% ± 3.82 UAR. While this result is lower than the strongest subject-wise audio-dominant benchmark, the framework contributes interpretability, modality analysis, AU dynamics, saliency-guided visual preprocessing, and ablation-based model selection.

VI. DISCUSSION

A. Interpretation of Overall Performance

The experimental results show that the proposed ESRA-RELA++ framework achieved its best performance using the calibrated simple-average fusion configuration without LoRA and without post-hoc specialist correction. This configuration achieved 72.40% ± 3.81 accuracy, 71.94% ± 4.12 macro-F1, and 72.67% ± 3.82 UAR under subject-wise 5-fold evaluation. The result indicates that the four active modalities—hybrid audio, saliency-guided facial video, AU statistical features, and AU-dynamic trajectory features—provide complementary information for emotion recognition.

A key observation from the experiments is that the best-performing model was not the most complex version of the framework. The final selected model used calibrated average fusion rather than reliability-aware meta-fusion, LoRA-based audio adaptation, pair specialists, or Sad-vs-rest correction. This suggests that, under the limited-size and subject-wise RAVDESS setting, simpler and more stable fusion can generalize better than heavily parameterized or highly corrected pipelines. In other words, adding more modules does not automatically improve unseen-actor performance. The model must not only learn emotional cues but also avoid overfitting to actor-specific vocal and facial characteristics.

The fold-wise results also show that performance varies across actor groups. Fold 1 achieved the highest accuracy of 77.33%, while Fold 3 achieved the lowest accuracy of 66.33%. This difference suggests that some actor groups are more difficult to generalize to than others. In subject-wise emotion recognition, variation in speaking style, facial expressiveness, gender, intensity of expression, and actor-specific articulation can strongly affect performance. Therefore, the standard deviation across folds is important because it reflects the stability of the model across unseen speakers and faces.

B. Why Calibrated Simple-Average Fusion Worked Best

One of the most important findings of this study is that calibrated simple-average fusion outperformed more complex fusion and correction mechanisms. The final selected configuration averaged the calibrated posterior probabilities from four active branches:

$$P_{final} = \frac{1}{4}(P_{audio} + P_{video} + P_{AU\text{-}stat} + P_{AU\text{-}dyn}). \quad (29)$$

TABLE X
FUSION-LEVEL CONFIGURATION COMPARISON

Configuration	LoRA	Specialists	Fusion Strategy	Accuracy	Macro-F1	UAR
ESRA-RELA++ final selected	No	No	Calibrated simple average	72.40%	71.94%	72.67%
ESRA-RELA++ full model without LoRA	No	Yes	Meta/dual fusion + specialists	71.48%	71.16%	71.77%
ESRA-RELA++ with LoRA	Yes	Yes	Meta/dual fusion + specialists	70.07%	68.56%	69.95%

This fusion rule is simple, but it has several advantages. First, it reduces the risk of overfitting because it does not introduce an additional high-capacity meta-classifier. Second, it treats all four modalities as complementary sources of information. Third, calibration helps reduce overconfidence from individual modality classifiers before averaging. As a result, no single unstable branch dominates the final decision.

The results suggest that the individual branches contain useful but imperfect information. The audio branch captures vocal and prosodic cues, the video branch captures facial appearance and selected expressive frames, the AU statistical branch captures global facial muscle activation, and the AU-dynamic branch captures temporal changes in facial expression. Since each modality has different strengths and weaknesses, averaging their calibrated predictions provides a balanced decision.

In contrast, the more complex meta-fusion and specialist-based configurations performed slightly worse. This may be because the training dataset is relatively small for learning a reliable second-level fusion model under subject-wise splitting. The meta-fusion model may learn patterns that work well for training actors but do not transfer equally well to unseen actors. Similarly, specialist modules may over-correct predictions for ambiguous samples. Therefore, the best result supports the idea that stable multimodal complementarity can be more effective than excessive post-processing complexity.

C. Analysis of LoRA-Based Audio Adaptation

LoRA was implemented as an optional parameter-efficient audio adaptation branch. The purpose of this branch was to allow the pretrained wav2vec2 model to adapt to emotion classification while updating only a small number of trainable parameters. This is theoretically attractive because full fine-tuning of large speech models can be expensive and may overfit on small datasets.

However, the experimental results showed that LoRA did not improve the final subject-wise performance in the current configuration. The LoRA-based model achieved lower overall performance than the no-LoRA configuration. The no-LoRA final selected model achieved 72.40% accuracy and 72.67% UAR, while the LoRA-enhanced configuration achieved lower performance in the full comparison.

This result does not mean that LoRA is generally ineffective for speech emotion recognition. Instead, it indicates that the current LoRA setup was not optimal for this specific dataset and evaluation protocol. RAVDESS contains only 1440 full audio-visual speech samples in the selected subset, and subject-wise splitting further reduces the amount of training data available in each fold. Under this condition, even parameter-efficient adaptation can overfit to training actors or fail to learn robust emotion-specific features. Another possible reason is that `facebook/wav2vec2-base-960h` is pretrained mainly for speech recognition, not emotion recognition. Without a stronger emotion-adapted speech backbone or careful validation-based hyperparameter tuning, the LoRA branch may produce noisy probability distributions that reduce final fusion quality.

Therefore, LoRA is best treated as an ablation in this study. Its inclusion is methodologically useful because it shows that parameter-efficient fine-tuning was tested under a leakage-safe protocol, but it was not selected for the final model because it did not improve subject-wise generalization.

D. Effect of Specialist Modules

The framework also included optional weak-pair specialists and a Sad-vs-rest specialist. These modules were designed based on earlier error patterns, where emotions such as Sad, Fearful, and Surprised were frequently confused. The weak-pair specialists targeted confusion pairs such as Fearful–Surprised, Sad–Disgust, Sad–Fearful, Angry–Disgust, and Neutral–Calm. The Sad-vs-rest specialist was introduced because Sad consistently appeared as one of the weakest classes.

Although these modules were conceptually useful, they were disabled in the final selected configuration because they did not improve the best observed performance. This indicates that post-hoc correction can be risky when the base fusion probabilities are already reasonably stable. A specialist classifier focuses only on a narrow decision boundary. If it is triggered incorrectly, it may shift probability mass away from the correct class and reduce overall performance. This is especially problematic in subject-wise evaluation, where the specialist may learn actor-dependent patterns from training folds that do not generalize well to unseen actors.

The failure of specialist modules to improve the final model is still an important research finding. It shows that targeted correction strategies should be validated carefully rather than assumed to be beneficial. In this study, the specialist modules are useful for error analysis and ablation discussion, but not for the selected final prediction pipeline.

E. Class-Wise Strengths and Weaknesses

The class-wise results show that the model performed best on Happy, Disgust, Calm, and Angry. These emotions often have stronger and more distinct facial or vocal patterns. For example, Happy and Disgust can involve clear facial muscle activations, while Angry may show strong vocal intensity and facial tension. The multimodal framework can capture these cues through a combination of audio, visual, and AU-based branches.

The weakest classes were Sad, Fearful, and Surprised. Sad achieved relatively low recall compared with the stronger classes. This may be because Sad expressions are often subtle and can overlap with low-arousal emotions such as Neutral or Calm. Sad speech may also have lower energy and slower prosody, which can be confused with calm or neutral speaking styles. On the visual side, Sad facial expressions may be less intense than emotions such as Happy, Angry, or Disgust.

Fearful and Surprised are also difficult to separate because they share similar high-arousal facial characteristics. Both may involve widened eyes, raised eyebrows, and mouth opening. Although AU-dynamic features help capture temporal changes, the overlap between these expressions remains challenging. This explains why high-arousal confusion pairs are common in multimodal emotion recognition.

These class-wise results show that the model is not uniformly weak. Instead, it performs well on emotions with distinct audio-visual patterns and struggles more with subtle or visually overlapping classes. This supports the need for future work on class-specific modeling, stronger temporal representation, and improved audio adaptation.

F. Importance of AU-Dynamic Features

A major novelty of ESRA-RELA++ is the inclusion of both AU statistical and AU-dynamic branches. Traditional AU-based approaches often summarize facial muscle activation using global statistics. While these statistics are useful, they may lose temporal information about how expressions develop during a clip.

The AU-dynamic branch addresses this by extracting segment-wise AU means, segment-wise maxima, AU velocity, peak timing, and FACS-group trajectory features. This design allows the model to capture not only whether an AU was active, but also when and how strongly it changed over time. This is important because facial expressions are dynamic processes rather than static images. For example, the timing of eyebrow raising, mouth opening, or cheek movement may help distinguish one emotion from another.

The final selected model includes AU-dynamic features as one of the four active branches. This supports the value of dynamic facial muscle representation in a multimodal emotion-recognition framework. However, the current AU-dynamic branch still uses fixed statistical trajectory features rather than a trainable temporal model such as a recurrent network or temporal transformer. Future work could improve this branch by learning AU sequences directly.

G. Comparison with Prior Work

The proposed framework does not surpass the strongest reported subject-wise RAVDESS result. For example, Luna-Jiménez *et al.* reported a higher subject-wise RAVDESS accuracy using a strong audio-based model combined with facial Action Units [2]. This difference can be explained partly by the strength of their audio branch. Their approach relies heavily on a highly optimized audio representation, whereas this work focuses on a broader explainable multimodal framework with audio, video, AU statistics, and AU dynamics.

Therefore, the contribution of this study should not be interpreted as achieving state-of-the-art accuracy. Instead, the contribution lies in the design and analysis of an explainable multimodal pipeline under strict subject-wise evaluation. The proposed framework investigates several important components: saliency-guided facial frame selection, hybrid audio features, AU statistical modeling, AU-dynamic modeling, calibrated fusion, LoRA adaptation, and specialist correction. The experiments show which components improve performance and which do not.

This is important because some prior works report high RAVDESS accuracy under different evaluation protocols. Trial-based or random splits may allow actor-specific characteristics to appear in both training and testing, making the task easier. In contrast, this study uses actor-independent folds, where test actors are unseen during training. As a result, the reported accuracy reflects a stricter and more realistic generalization setting.

H. Analytical Insight from the Ablation Results

The ablation results provide one of the strongest analytical contributions of this work. They show that the best-performing configuration was not the most complex one. The LoRA-enhanced model and the specialist-corrected model did not outperform the simpler calibrated average fusion model.

This result suggests three important points. First, model complexity must be justified empirically. Adding trainable adaptation layers or post-hoc correction modules can increase the risk of overfitting, especially when the dataset is small and evaluation is subject-wise. Second, branch quality matters more than branch quantity. If an additional branch produces weak or noisy predictions, the fusion model may degrade even if the branch is theoretically meaningful. This was observed with the LoRA branch under the current setup. Third, simple fusion can be strong when branches are complementary. The calibrated average fusion performed best because the four active branches provided diverse but stable information. This supports the value of modular multimodal design with careful ablation.

I. Challenges Identified During Experimentation

Several challenges were identified during the development and evaluation of the system. The first challenge was dataset filtering. The local RAVDESS folder contained 4904 video files, but not all of them were suitable for the proposed audio-visual speech task. The model required full audio-video speech files, so video-only and song files were excluded. This reduced the dataset to 1440 full-AV speech samples. Although this made the experiment cleaner and more comparable, it also limited the amount of training data.

The second challenge was OpenFace preprocessing. The proposed framework depends on OpenFace CSV files for AU features, facial landmarks, saliency selection, and reliability information. Missing or failed OpenFace outputs could weaken the visual and AU branches. In the final experiment, OpenFace coverage was complete, which strengthened the reliability of the preprocessing pipeline.

The third challenge was actor-independent generalization. Subject-wise splitting is more difficult than random splitting because the model must classify emotions from unseen actors. The performance gap across folds shows that actor identity and expression style strongly affect generalization. The fourth challenge was weak-class recognition. Sad, Fearful, and Surprised remained difficult even after adding AU-dynamic features and testing specialist correction. This indicates that some emotion pairs require stronger temporal or semantic modeling than fixed feature extraction can provide. The fifth challenge was balancing complexity and generalization. LoRA and specialist modules increased architectural complexity but did not improve the final selected result. This confirms that complex modules must be evaluated carefully in small subject-wise datasets.

J. Limitations

This study has several limitations. First, the final accuracy is lower than the strongest reported subject-wise RAVDESS benchmark. Therefore, the proposed method should be presented as an explainable and modular framework rather than a new state-of-the-art accuracy model.

Second, the audio branch is not as strong as highly optimized fine-tuned audio models. The final selected model uses frozen wav2vec2 embeddings and handcrafted features. Although LoRA was tested, it did not improve the final result under the current setup. Future work should explore stronger audio backbones such as XLSR-Wav2Vec2 or WavLM, emotion-pretrained speech models, or carefully validated full fine-tuning.

Third, the video branch uses MobileNetV3 frame embeddings with temporal pooling rather than a trainable temporal video model. This lightweight design reduces overfitting risk but may miss complex facial motion patterns. A stronger temporal module, such as a BiLSTM, temporal attention, or transformer encoder, may improve visual emotion representation.

Fourth, the AU-dynamic branch uses engineered temporal statistics rather than learned AU sequence modeling. Although this improves interpretability, it may not fully capture complex temporal dependencies. Future work could train a lightweight AU sequence model using actor-safe validation. Finally, the study uses only RAVDESS full audio-visual speech samples. The findings should be validated on additional datasets to test cross-dataset generalization.

K. Summary of Discussion

Overall, the discussion shows that ESRA-RELA++ provides a meaningful and explainable multimodal framework for subject-independent emotion recognition. The best result was achieved by calibrated simple-average fusion across four active branches: hybrid audio, saliency-guided video, AU statistics, and AU dynamics. More complex modules, including LoRA audio adaptation and specialist correction, did not improve the final selected performance under the current setting. This finding is important because it demonstrates that reliable generalization in small subject-wise datasets depends not only on architectural novelty, but also on stability, calibration, and careful ablation.

The proposed system should therefore be positioned as an interpretable multimodal framework rather than a purely accuracy-driven model. Its main strengths are modularity, saliency-guided preprocessing, AU-dynamic modeling, subject-wise evaluation, uncertainty analysis, and transparent ablation-based model selection.

VII. Conclusion and Future Work

A. Conclusion

This paper presented ESRA-RELA++, an explainable saliency and reliability-aware multimodal framework for subject-independent emotion recognition using the RAVDESS full audio-visual speech subset. The proposed system was designed to combine complementary emotional cues from four active modalities: hybrid audio features, saliency-guided facial video features, Action Unit statistical features, and AU-dynamic trajectory features. The system was evaluated using a strict subject-wise 5-fold protocol, where test actors were completely unseen during training. This evaluation setting is more challenging than random or trial-based splitting because the model must generalize to new speakers and faces rather than relying on actor-specific patterns.

The final selected configuration used calibrated simple-average fusion across the four active branches. This configuration achieved 72.40% $\pm$ 3.81 accuracy, 71.94% $\pm$ 4.12 macro-F1, and 72.67% $\pm$ 3.82 UAR under subject-wise 5-fold evaluation. The final setup used the audio, video, AU-statistical, and AU-dynamic branches, while LoRA audio adaptation, pair-specialist correction, and Sad-vs-rest correction were disabled because they did not improve the best observed performance.

The results show that multimodal features provide useful complementary information for emotion recognition. The audio branch captured vocal and prosodic patterns, the video branch captured saliency-guided facial appearance, the AU-statistical branch represented global facial muscle behavior, and the AU-dynamic branch captured temporal facial expression changes. Among the emotion classes, Happy, Disgust, Calm, and Angry were recognized more effectively, while Sad, Fearful, and Surprised remained more challenging. This indicates that emotions with stronger and more distinct audio-visual patterns are easier to classify, while subtle or visually overlapping emotions require more advanced modeling.

A key finding of this work is that the most complex model was not the best-performing model. The LoRA-enhanced version and specialist-corrected versions did not outperform the calibrated simple-average fusion configuration. This suggests that, for a relatively small actor-independent dataset such as RAVDESS full audio-visual speech, additional trainable modules or post-hoc correction mechanisms may introduce noise or overfitting. The best result was obtained by using stable complementary modality predictions and combining them through a simple calibrated fusion rule. This finding supports the importance of careful ablation and model selection rather than assuming that architectural complexity always improves performance.

Although the proposed model does not exceed the strongest previously reported subject-wise RAVDESS accuracy, its contribution lies in the design of a modular, interpretable, and analytically transparent multimodal framework. The work provides detailed branch-wise experimentation, LoRA/no-LoRA comparison, specialist-module analysis, uncertainty logging, and class-wise performance interpretation. Therefore, ESRA-RELA++ is positioned not only as an emotion-recognition model, but also as an explainable experimental framework for studying how audio, facial appearance, facial Action Units, and AU dynamics contribute to subject-independent emotion recognition.

B. Main Findings

The major findings of this study can be summarized as follows. First, subject-wise evaluation is difficult but important. The use of unseen actors in the test folds makes the task more realistic and challenging. Fold-wise variation showed that actor identity, expressiveness, speaking style, and facial behavior strongly affect generalization.

Second, four-branch multimodal fusion improved interpretability. The final framework combined hybrid audio, saliency-guided video, AU statistics, and AU dynamics. This allowed the model to use both vocal and facial information while keeping the contribution of each branch analyzable.

Third, calibrated simple-average fusion worked better than complex fusion. The best-performing configuration used simple calibrated averaging rather than meta-fusion, LoRA adaptation, or specialist correction. This shows that stable fusion can be more effective than excessive post-processing on small subject-wise datasets.

Fourth, LoRA did not improve performance in the current setup. LoRA was tested as a parameter-efficient audio adaptation branch, but the LoRA-enhanced configuration performed worse than the no-LoRA selected model. Therefore, LoRA was retained as an ablation rather than used in the final model.

Fifth, Sad, Fearful, and Surprised remained difficult classes. These emotions showed lower recall compared with stronger classes such as Happy, Disgust, Calm, and Angry. This suggests that subtle low-arousal emotions and high-arousal overlapping emotions require stronger temporal and class-specific modeling.

Finally, AU-dynamic features added meaningful interpretability. The AU-dynamic branch captured segment-wise AU behavior, AU velocity, peak timing, and FACS-group trajectories. Even though the final model used simple fusion, the AU-dynamic branch helped make the framework more explainable and facial-behavior aware.

C. Limitations

Despite the promising structure of the proposed framework, several limitations remain. First, the final accuracy is lower than the strongest reported subject-wise RAVDESS benchmark. The proposed system should therefore not be presented as a state-of-the-art accuracy model. Instead, it should be interpreted as an explainable and modular multimodal framework.

Second, the audio branch is still weaker than highly optimized audio-dominant systems. The final model used frozen wav2vec2 embeddings with handcrafted acoustic-prosodic features. Although LoRA was evaluated, it did not improve the final performance under the current configuration. This suggests that stronger audio adaptation is needed.

Third, the video branch used MobileNetV3 frame embeddings with temporal pooling. While this design is lightweight and reduces overfitting risk, it may not fully capture complex facial motion patterns across time. Fourth, the AU-dynamic branch used engineered temporal statistics rather than a trainable sequence model. This improves interpretability but may limit the ability to learn deeper temporal dependencies in facial muscle movement.

Finally, the study was performed only on the RAVDESS full audio-visual speech subset. The conclusions should be validated on additional datasets before claiming general robustness across different recording conditions, languages, cultures, or spontaneous emotions.

D. Future Work

Future work can improve the proposed system in several directions.

1) Stronger Audio Adaptation: The largest performance gap compared with stronger prior work appears to be the audio branch. Future experiments should explore stronger audio models such as XLSR-Wav2Vec2, WavLM, or emotion-pretrained speech models. Instead of using only frozen embeddings, future work may apply carefully validated fold-safe fine-tuning or parameter-efficient adaptation. However, any audio adaptation must preserve the subject-wise protocol so that test actors are never used during training or model selection.

2) Improved LoRA Configuration: Although LoRA did not improve the current result, it may still be useful with better settings. Future work can test different LoRA ranks, learning rates, target modules, epoch schedules, and audio backbones. For example, applying LoRA to an XLSR or WavLM backbone may produce stronger emotion-specific audio representations than using `facebook/wav2vec2-base-960h`. LoRA should continue to be evaluated as an ablation before being included in the final model.

3) Temporal Video Modeling: The current visual branch uses saliency-selected frames and temporal pooling, but it does not train a dedicated temporal video model. Future work could add a lightweight temporal model such as a BiLSTM, temporal attention, 1D temporal CNN, or small transformer encoder. This may help the system learn how facial expressions evolve across time rather than relying only on pooled frame-level features.

4) AU Sequence Learning: The AU-dynamic branch currently uses fixed temporal statistics. A future version could train a small AU sequence model using frame-level AU vectors. For example, OpenFace AU sequences could be passed into a BiLSTM or temporal attention model to learn facial muscle movement patterns directly. This may improve difficult emotion pairs such as Fearful–Surprised and Sad–Neutral.

5) Better Weak-Class Handling: Sad, Fearful, and Surprised remained the weakest classes. Future work should focus on class-specific improvement strategies, such as Sad-vs-rest modeling with stronger validation, Fearful–Surprised temporal separation, class-balanced loss functions, emotion-specific augmentation, and threshold tuning using only validation folds. However, post-hoc correction should be applied carefully because the specialist modules in this study did not improve the final selected result.

6) Cross-Dataset Validation: The proposed framework should be tested on additional audio-visual emotion datasets to evaluate generalization beyond RAVDESS. Cross-dataset experiments would show whether the learned multimodal strategy is robust to different speakers, recording settings, languages, and expression styles.

7) Explainability and Error Analysis: Future work can extend the uncertainty and error-analysis components. For example, low-confidence samples could be analyzed visually and acoustically to identify why the model fails. AU activation plots, saliency-frame visualizations, confusion-pair analysis, and per-actor performance analysis could make the framework more interpretable.

8) Lightweight Deployment: Since the final selected model uses simple calibrated average fusion and fixed feature extraction, it may be suitable for lightweight deployment after preprocessing. Future work could investigate model compression, faster OpenFace alternatives, real-time feature extraction, and edge-device inference.

E. Final Statement

In conclusion, ESRA-RELA++ provides a structured and explainable multimodal framework for subject-independent emotion recognition on RAVDESS. The study demonstrates that audio, facial video, AU statistics, and AU dynamics provide complementary information, but also shows that additional complexity such as LoRA adaptation and specialist correction does not always improve generalization. The final selected calibrated simple-average fusion model achieved 72.40% accuracy, 71.94% macro-F1, and 72.67% UAR, offering a stable and interpretable baseline for future research in subject-independent multimodal emotion recognition.

REFERENCES

[1] S. R. Livingstone and F. A. Russo, "The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS): A dynamic, multimodal set of facial and vocal expressions in North American English," *PLOS ONE*, vol. 13, no. 5, Art. no. e0196391, 2018, doi: 10.1371/journal.pone.0196391.

[2] C. Luna-Jiménez, R. Kleinlein, D. Griol, Z. Callejas, J. M. Montero, and F. Fernández-Martínez, "A proposal for multimodal emotion recognition using aural transformers and action units on RAVDESS dataset," *Applied Sciences*, vol. 12, no. 1, Art. no. 327, 2022, doi: 10.3390/app12010327.

[3] V. John and Y. Kawanishi, "Audio and video-based emotion recognition using multimodal transformers," in *Proc. 26th International Conference on Pattern Recognition (ICPR)*, 2022, pp. 2582–2588, doi: 10.1109/ICPR56361.2022.9956730.

[4] A. Baevski, H. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 12449–12460.

[5] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," arXiv:2106.09685, 2021.

[6] P. Ekman, W. V. Friesen, and J. C. Hager, *Facial Action Coding System: The Manual on CD ROM*. Salt Lake City, UT, USA: A Human Face, 2002.

[7] T. Baltrušaitis, A. Zadeh, Y. C. Lim, and L.-P. Morency, "OpenFace 2.0: Facial behavior analysis toolkit," in *Proc. 13th IEEE International Conference on Automatic Face & Gesture Recognition (FG)*, 2018, pp. 59–66, doi: 10.1109/FG.2018.00019.

[8] A. Howard, M. Sandler, G. Chu, L.-C. Chen, B. Chen, M. Tan, W. Wang, Y. Zhu, R. Pang, V. Vasudevan, Q. V. Le, and H. Adam, "Searching for MobileNetV3," in *Proc. IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 1314–1324.

[9] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019.

[10] T. Wolf *et al.*, "Transformers: State-of-the-art natural language processing," in *Proc. 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, 2020, pp. 38–45.

[11] B. McFee *et al.*, "librosa: Audio and music signal analysis in Python," in *Proc. 14th Python in Science Conference (SciPy)*, 2015, pp. 18–25.

[12] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[13] G. Bradski, "The OpenCV library," *Dr. Dobb's Journal of Software Tools*, 2000.